GlaucoScan — RE-GAN Analysis Report

Fundus Image Preprocessing & Clinical Feature Extraction

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11.3%

Low Risk Optic nerve head appears within normal limits. Routine annual eye examination recommended.

Preprocessing Results

The fundus image was processed through a 5-stage classical pipeline followed by RE-GAN deep learning enhancement. The three images below show the progression from raw input to fully enhanced output.

Original (Raw input)

Classical Pipeline (Stages 1–5)

RE-GAN Enhanced (Deep learning)

Classical pipeline: 30.0 ms RE-GAN enhancement: 2131.3 ms Total: 2161.3 ms

Extracted Clinical Features

The following five biomarkers were extracted from the RE-GAN-enhanced image using the two-stage optic disc localisation algorithm. All features are grounded in published ophthalmological literature.

Feature	Value	Clinical Range	Interpretation	Weight
Vertical CDR	0.367	< 0.65 normal	Normal	50%
Horizontal CDR	0.544	< 0.65 normal	Borderline	—
Rim-Disc Area Ratio	0.782	> 0.50 normal	Normal rim tissue	25%
ISNT Score	3 / 3	3 = fully normal	Full compliance (normal)	15%
Vessel Density	0.62%	> 10% normal	Reduced vessel density	10%

Optic Disc & Cup Measurements

Measurement	Value	Measurement	Value

Disc Height (px)	159.4	Cup Height (px)	58.4
Disc Width (px)	81.0	Cup Width (px)	44.1
Disc Centre X	110	Image Sharpness	19.53
Disc Centre Y	109	Enhancement	RE-GAN (U-Net+CBAM)

Glaucoma Risk Score Breakdown

The overall glaucoma probability of 11.3% is computed using a weighted clinical scoring formula based on published ophthalmological thresholds. Each feature contributes independently to the final score.

Feature	Raw Score	Weight	Contribution
Vertical CDR	5%	50%	2.5%
Rim-Disc Area Ratio	5%	25%	1.2%
ISNT Score	0%	15%	0.0%
Vessel Density	75%	10%	7.5%
TOTAL		**100%**	**11.3%**

Preprocessing Pipeline Used

Stage	Operation	Purpose	Status
1	Green Channel Extraction	Maximise disc/vessel contrast	✓ Applied
2	Gaussian BG Normalisation	Remove non-uniform illumination (Novel)	✓ Applied
3	CLAHE Enhancement	Boost local contrast	✓ Applied
4	Bilateral Denoising	Edge-preserving noise removal	✓ Applied
5	Gamma Correction	Brightness adjustment	✓ Applied
6	RE-GAN Enhancement	Deep learning enhancement (U-Net+CBAM)	✓ Applied

Important Notice

This report is generated by an automated research system (GlaucoScan RE-GAN) developed as part of a PhD research programme. It is intended for research and educational purposes only and does NOT constitute a medical diagnosis. All findings must be confirmed by a qualified ophthalmologist before any clinical decision is made. If you are experiencing vision problems, please consult a healthcare professional immediately.